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Importance measures in risk-informed decision making: Ranking, optimisation and configuration control

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ABSTRACT

This paper describes roles, extensions and applications of importance measures of components and configurations for making risk-informed decisions relevant to system operations, maintenance and safety. Basic importance measures and their relationships are described for independent and mutually exclusive events and for groups of events associated with common cause failures. The roles of importances are described mainly in two groups of activities: (a) ranking safety significance of systems, structures, components and human actions for preventive safety assurance activities, and (b) making decisions about permissible permanent and temporary configurations and allowed configuration times for regulation, technical specifications and for on-line risk monitoring. Criticality importance and sums of criticalities turn out to be appropriate measures for ranking and optimization. Several advantages are pointed out and consistent ranking of pipe segments for in-service inspection is provided as an example.

Risk increase factor and its generalization risk gain are most appropriately used to assess corrective priorities and acceptability of a situation when components are already failed or when planning to take one or more components out of service for maintenance. Precise definitions are introduced for multi-failure configurations and it is shown how they can be assessed under uncertainties, in particular when common cause failures or success states may be involved. A general weighted average method is compared to other candidate methods in benchmark cases. It is the preferable method for prediction when a momentary configuration is known or only partially known. Potential applications and optimization of allowed outage times are described.

The results show how to generalize and apply various importance measures to ranking and optimization and how to manage configurations in uncertain multi-failure situations.

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1. Introduction

Methodologies for risk-informed decision making categories have been tested and applied in recent decades in various industries. The principal driving force for risk assessment applications is the need to optimize or prioritize objects for betterment of plant systems, operations, safety management and preventive or corrective maintenance activities [1]. The outcome is improved economy and effective use of resources while maintaining or improving safety. Some of the methods have even been established to the level of Regulatory Guides in nuclear industry [2–5]. Reviews of typical applications have pointed out that importance measures can play significant roles in these activities [6–8]. However, according to some reviews of issues in risk informed applications of probabilistic safety assessment (PSA) there is confusion and lack of consensus as to whether *common cause failure* (CCF) events should be included when assessing importance measures for systems, components and structures, and how to treat them, and how to map basic event importances to component or system importances [9]. Ref. [10] identified subjects demanding further development in risk monitors, e.g. CCF modeling when reductions in redundancy occur when components are removed from service or failures are identified, and ways of handling negations in PSA. Somewhat ad hoc methods have been used in the past to handle these problems. The present paper clarifies and develops solutions for several of these issues. It is neither possible nor necessary to identify one general importance measure that is the correct one for all applications. Different application categories need different measures, each serving the specific objective of an application.

Basic importance measures are defined in Section 2 consistent with early Ref. [11], extended to system models that may contain groups of mutually exclusive (m.e.) basic events in addition to more commonly assumed statistically independent events. Through Boolean algebra this has implications to components that have multiple failure modes or participate in CCFs. The relationships and additivity of importance measures depend on these properties.

One objective of risk-informed regulation and management is to optimize or prioritize components for actions and resources

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available for improved safety and economy. Section 3 is devoted to ranking safety significance of systems, structures, components and human actions (SSCH) for preventive safety assurance activities (PSAA) that are expected to improve reliability characteristics of SSCH. Such activities include safety classification, testing and inspections, quality assurance, preventive maintenance, training, procedures and human factors development. A human factors program may consist of actions developing personnel training, ergonomics, procedures and practices that influence human error probabilities in a risk model. It is obvious that to have sensible ranking one must select for each SSCH a set of basic events that are sensitive to a particular PSAA and are significant for risk.

It has been claimed earlier [12] that only a recently introduced differential importance measure (DIM) [13] can be used for ranking both process pipes (failures of which cause initiating events) and safety system pipes (failures of which cause unavailabilities of safety components) consistently for in-service inspection planning. Section 3.1 illustrates with an example that criticality importance (CI) can be used equally and yields identical ranking with a common scale for process pipes and safety system pipes. Section 3.2 shows how the sums of criticalities over certain groups of basic events are important characteristics for measuring impacts of multiple changes in reliability characteristics or inspection intervals. Cost-rate minimization by optimizing test and maintenance intervals is another topic in which the criticality importance plays an important role. These properties make CI the most appropriate measure for ranking and categorization.

The subject of Section 4 is risk-informed configuration management in a broad sense. It applies to decision making about plant modifications and allowed durations of repairs or temporary configurations. The terminology of "configuration" is carefully defined for momentary and predictive purposes and for cases of uncertainty about which basic events have actually occurred or what has to be assumed for the future based on momentary information at hand. A risk gain (RG) is defined as a generalized risk increase factor for a configuration, and this measure can be larger than 1 but also smaller than 1 if success states can be justified. It is shown that even if negations appear with success states in exact formulas, all problems can be treated with conventional software tools. Several candidate methods are compared for handling uncertain configurations and CCF and for combining basic event information to a component or to a system, and are shown to be equal under special configurations but not always. The weighted average method (WAM) seems to be correct for most general uncertain and multi-component configurations. Numerical examples are intentionally kept simple to make sure that key issues and findings are clearly demonstrated. A concrete example illustrates quantification of RG for components participating in CCF and having several failure modes. A practical optimization procedure for balancing optimal allowed outage times is presented in Appendix A. Section 5 includes a summary and final conclusions. This paper is an update and extension of works published earlier in conferences [14,15].

1.1. Notation and definitions

$Y=Y(Z_1, Z_2, \ldots, Z_K)$ TOP-event of a fault tree; any Boolean function of basic events Z_k

$X=X(Z_1, Z_2, \ldots)$ configuration, a Boolean expression of what is known or assumed to be failed or unfailed. *A* configuration *X* can be based on real knowledge at some moment, or it can be a predictive situation contemplated in "what if" questions when planning future maintenance actions or technical specifications

$P(Y|X)=P(Y|X=\text{TRUE})$ conditional probability (or rate) of *Y*, given configuration *X*

$P(Y|X)$ $P(XY)/P(X)$

$z_k=P(Z_k)$ probability of a basic event Z_k, or frequency of an initiating event

$y=P(Y)=y(z_1, z_2, \ldots, z_K)$ probability of the TOP event; accident rate

k index of Z_k or z_k. Each index set *k* must indicate, which components failed, one or more in a common cause event (CCF), and the failure mode (if more than one) and possibly the plant operating state. For example, $Z_{1,2;\ a}$ indicates a CCF cause event failing components 1 and 2 simultaneously in failure mode a

$x=P(X)=x(z_1, z_2, \ldots)$ probability of a configuration

Z' logic negation of any Boolean variable *Z* (NOT-*Z*)

XZ logic product (AND-gate) of any binary (Boolean) events *X* and *Z*

$X+Z$ logic sum (OR-gate) of any binary Boolean events *X* and *Z*

ACT allowed configuration time

AOT allowed outage (repair) time

CCF common cause failure

MCS minimal cut set

m.e. mutually exclusive

m.s.i. mutually statistically independent

POS plant operating state (full power, cold shutdown, etc.)

PSAA preventive safety assurance activities

RIF risk increase factor (equal to risk achievement worth, RAW)

s.i. statistically independent

SSCH systems, structures, components and human actions

2. Importance measures

Different application categories employ different kinds of importance measures. We start with somewhat generalized definitions of risk importances to clarify issues relevant to mutual exclusivity and independence. Most importance measures were originally defined and applied for independent (s-independent) basic events. This continued even when at the same time it was claimed that some events were mutually exclusive. Here mutually exclusive and mutually independent events are explicitly treated. The distinction is particularly important when a component can have multiple failure modes or is a member of a common cause component group subject to one or more common failure causes. It will be shown how the concepts influence complex decision making situations expressed by Boolean functions.

The following generally valid rules explain some of the mathematical developments that follow: for any Boolean variables or functions *G* and *H* holds $G+GH=G$, $G+H=GH'+H$, $P(G+H)=P(G)+P(H)-P(GH)$ and $P(GH')=P(H)-P(GH)$.

2.1. Mutually exclusive events and mutually statistically independent events

Consider a system in which some sets of basic events are mutually statistically independent (m.s.i.) and other sets are mutually exclusive (m.e.). Any event can be m.s.i. with some events and m.e. with others, and no other kinds of events are assumed. The m.s.i. between Z_i and Z_j means that $P(Z_iZ_j)=P(Z_i)P(Z_j)$, and m.e. means that $Z_iZ_j=\phi$ (an impossible event; empty set), which implies $P(Z_iZ_j)=0$. No two real events can be both m.e. and m.s.i. simultaneously. Mutual exclusivity means that if Z_i is set TRUE ($z_i=1$) then Z_j is FALSE ($z_j=0$). However, to set some Z_i FALSE ($z_i=0$) does not cause any change in other event probabilities in our applications. It is typical that if one failure mode of

a component is eliminated, the other modes can still occur with the original probability but do not occur with certainty.

Consider a system with a subset J of basic events satisfying the following conditions:

- All distinct events Z_j of J are m.e.
- Events are m.s.i. unless they are m.e.
- Any basic event Z_j of J is m.s.i. with all events outside J, or at least with events in the same minimal cut sets (MCS) with Z_j and in MCS that contain no member of J.

Evidently no MCS can contain more than one member of the same J. The definition allows a member of J to be m.e. with an event outside J when this is a factor in a MCS with another member of J. Despite the abstract nature, this kind of subset J could be all basic events that fail one specific component, either single failure modes alone or including both single failures and relevant CCFs. Also the initiating events of a risk model are usually considered mutually exclusive and form a set J. A single independent basic event can be the smallest set J. A system can have multiple subsets that satisfy the above conditions. This kind of subset is important when considering ranking and configuration control of components with multiple failure modes and common cause failures. This will be illustrated with examples later.

Consider J defined by the index set $j=1, 2, \ldots, J$. The fault tree TOP event can be written generally

$$Y = \sum_{j=1}^{J} Z_j G_j + H, \tag{1}$$

where Z_jG_j=union of the minimal cut sets (mcs) that contain Z_j as a factor, and H=union of the minimal cut sets that contain no $Z_j j \in J$. G_j and H contain no Z_j ($j \in J$), and $Z_jZ_k=\phi$ for all $j \neq k$, $j, k=1, 2, \ldots, J$.

The nominal risk y (risk at nominal values z_k) can be obtained with fault trees, Boolean algebra and probability theory. It equals

$$y = P(Y) = P(H) + \sum_{j=1}^{J} z_j P(G_j H'). \tag{2}$$

Next we consider some quantities essential for the importance measures related to basic event Z_k, $k \in J$. The low-end risk $y_k^- = y$(at $z_k=0$) can be obtained simply by dropping from Eq. (2) the terms proportional to z_k. It is equal to $y_k^- = P(Y|Z_k') = P(YZ_k')/P(Z_k') = [y - P(YZ_k)]/(1-z_k)$.

Criticality importance is defined as the term proportional to z_k divided by the total risk y

$$CI_k = z_k P(G_k H')/y = (y - y_k^-)/y. \tag{3}$$

This is easily obtained after y and y_k^- have been solved. The Risk Reduction Worth is $RRW_k = y/y_k^- = 1/(1-CI_k)$, i.e. $CI_k = (y - y_k^-)/y = 1 - RRW_k^{-1}$. Obviously any ranking based on the values $\{RRW_j\}$ is equivalent with ranking based on $\{CI_j\}$. From the definition follow important interpretations:

- CI_k is the fractional risk reduction if event k can be eliminated ($z_k \to 0$).
- CI_k is the fractional risk increase if the failure probability is doubled ($z_k \to 2z_k$).

Thus, CI_k immediately tells what happens to the risk if test interval for Z_k is doubled or halved, or if an improved preventive maintenance program reduces the failure rate (and z_k) by a certain fraction. CI_k is often approximated with the Fussell–Vesely importance (FV_k) defined in the rare-event approximation $\approx z_k P(G_k)/y$. [11].

The Birnbaum importance $BI_k = \partial y/\partial z_k = P(G_k H')$ does not depend on z_k but has interesting roles and benefits in calculating failure intensities, estimating reliabilities of repairable systems, and in estimating importances of uncertainties of input probabilities [16,25].

The high-end risk value y_k^+ at $z_k=1$ is $y_k^+ = P(Y|Z_k) = P(YZ_k)/P(Z_k) = P(G_k+H) = P(H) + P(G_kH')$, and the risk-increase factor is defined

$$RIF_k = y_k^+ / y. \tag{4}$$

It follows from these that the full system model is not needed to solve y_k^+ after y and the criticalities have been solved because

$$RIF_k = 1 + \frac{CI_k}{z_k} - \sum_{j=1}^{J} CI_j. \tag{5}$$

The traditional relationship for independent events is obtained with Z_k as the only member of set J so that the last sum of criticalities in Eq. (5) is CI_k. The reason for the sum in Eq. (5) is that the mutual exclusivity makes other events Z_j in J impossible when one Z_k occurs.

Mutual exclusivity is often assumed for sets of events that fail one or more common components (CCF) and for different single-failure modes of a component. A simple example is a two-component system with basic events Z_1, Z_2 and $Z_{1,2}$ such that the single failures Z_1 and Z_2 are m.s.i. while $Z_{1,2}$ (failing both components) is m.e. with both Z_1 and Z_2. The system model is $Y = Z_1Z_2 + Z_{1,2}$, and one subset J is the pair $(Z_1, Z_{1,2})$. One can easily verify Eq. (5) when $y = z_1z_2 + z_{1,2}$ and $RIF_1 = z_2/y$, $RIF_2 = z_1/y$, $RIF_{1,2} = 1/y$, and $CI_1 = CI_2 = z_1z_2/y$, $CI_{1,2} = z_{1,2}/y$. If one assumes mutual independence between all basic events (as conventional fault tree software usually does), then $y = z_1z_2 + z_{1,2} - z_1z_2z_{1,2}$ and all importance measures are somewhat different. Even with the rare-event approximation (first order Poincaré inclusion–exclusion formula) RIF_1 and RIF_2 differ from those obtained under the mutual exclusivity assumption. In this example one could just as well select the pair $(Z_2, Z_{1,2})$ as subset J.

The expression for BI_k is completely independent of z_k and RIF_k is nearly independent of the property z_k of the component whose importance is supposed to be measured, and what PSAA is supposed to influence. On the contrary, CI_k is very sensitive to z_k. This leads to very different applications for these importance measures.

2.2. Failure modes with similar minimal cut sets

Basic events Z_j and Z_k have "similar" minimal cut sets if all other basic events in their cut sets are the same, i.e. all cut sets of Z_j can be obtained by replacing Z_k by Z_j,and vice versa. The union of such events can be defined as a module. This can happen when Z_j and Z_k are failure modes of one component so that either one makes the component unavailable. For example, a leak of coolant of certain size class can consist of many pipeline events defined separately. Also a pump can have several piece-part failures. Common to all such situations is that each failure alone fails the unit as if all those failures occurred. So, RIF is the same for all events of a module. Because the sum of criticalities plays an important role in some applications, this subsection is to study the additivity of criticalities within a module.

A fault tree TOP event can be written in this case as follows:

$$Y = Z_1G + Z_2G + \ldots + Z_JG + H = ZG + H, \tag{6}$$

where the module Z is the union of the individual basic events Z_k, and G and H are assumed to be independent of all Z_k, $k=1, 2, \ldots, K$. Consider two cases:

a) *Basic events Z_j are m.e., $j=1, 2, \ldots, J$*: In this case the set $\{Z_j\}$ has the properties of set J of the previous Section and the Eqs. (1)–(5) apply. The individual importances BI_j and RIF_j are all the same and the same as for the single module event Z, which has the additive probability $z = \sum_{j=1}^{J} z_j$: $BI_j = BI(Z) = P(GH')$, and

$RIF_j = RIF(Z) = P(G+H)/y$, for all $j=1, 2, \ldots, J$. Furthermore, the criticality of the module Z is the sum of the individual criticalities: $CI(Z) = \sum_{j=1}^{J} CI_j$. Thus, both z and the criticality are additive. One can subdivide a pipeline to segments, define basic events for the parts, and the importance of each segment becomes a proper fraction proportional to the length (or to the number of welds, as appropriate). Or one can combine events into modules and simply add the probabilities and criticality importances.

b) *Basic events Z_j are m.s.i.*, $j=1, 2, \ldots, J$: The RIF_j values are equal to $RIF(Z)$. The Birnbaum importance for the module Z is again $BI(Z)=P(GH')$, but this is not the same for each individual Z_j. $P(Y)$ is $y=P(H)+zP(GH')$ with $z=[1-\prod_{j=1}^{J}(1-z_j)]$. Derivation yields $BI_k = BI(Z)\prod_{j=1, j\neq k}^{J}(1-z_j)$. Each CI_j is still roughly proportional to z_j, and the sum $\sum_{j=1}^{J} CI_j$ nearly equals the module criticality $CI(Z)=zBI(Z)/y$ when probabilities are small. In this case the subdivision partitions almost correctly the probability and the importance when individual probabilities z_k are small.

3. Ranking components for preventive activities

The objective of *preventive* activities (PSAA) is to influence risk through the reliability characteristics z_k of components and other objects (SSCH), recognizing that PSAA can make a difference in z_k but does not make the object perfect. Thus, the importance measure to be used in ranking must be sensitive to small changes in z_k. The importance must also tell how strongly that change influences risk, because the objective of ranking is to prioritize PSAA-efforts to SSCH that are most influential to risk. The criticality importance CI satisfies these requirements while the Birnbaum measure and RIF do not because they are insensitive to modest changes in event probabilities and measure the influence of complete failures.

The risk function of a plant can generally be presented as the accident frequency, a function

$$y = y(z_1, z_2, \ldots, z_{m+n}) = \sum_{i=1}^{m} f_i P_i(q_1, q_2, \ldots, q_n), \tag{7}$$

of probabilities $z_k=P(Z_k)$ of basic events Z_k. The probabilities are either initiating event rates f_i (probabilities per unit time), or dimensionless unavailabilities q_j, probabilities of failed states of safety components or probabilities of human errors. The dimension of y is per unit time or operating cycle. P_i is the conditional probability of an accident, given initiating event i. An accident is defined separately in each application. In nuclear industry y is often a reactor core damage frequency (CDF) or a large early radioactive release frequency (LERF). A complete model for y includes all initiating events, system models and basic events for all plant operating states covering the normal full power operating period, and periodic outages that can have several stationary states and transitions in shutting down and in starting up again. The total risk is calculated for an operating cycle of length L (typically a year), and the initiating event rates are defined per phase or cycle time.

In many cases PSAA can influence both initiating event rates (f_i) and safety system unavailabilities (q_j). Ranking must be possible consistently for both. This makes BI and RIF unsuitable for ranking because they have a different dimension or meaning for initiating events and unavailabilities.

There is another requirement for a good ranking importance measure. A component may be modeled by more than one failure mode, and a maintenance program or a changed inspection interval may affect all or several basic events. The importance measure should then be additive with respect to such events, as criticality usually is.

3.1. Ranking pipe segments for in service inspection

In-service inspection (ISI) is a practice to inspect periodically segments of piping to detect possible degradation and needs to repair or replace pipelines. The frequencies of inspections can be based on risk-importances of different segments. When planning inspection scheduling the methods should take into account failure mechanisms and vulnerabilities of piping segments as well as the risk-impacts of the pipe failures [5,17]. It is of interest to examine, which importance measures may be used in prioritizing inspection frequencies. It will be shown that criticalities CI_k can be used for ranking consistently both process system and safety system components.

The initiating event in this example of Ref. [12] is a medium size loss of coolant accidents (MLOCA) with total frequency $f=1.14\times10^{-3}$/yr, subdivided into 12 segments with frequencies f_i, $i=1, 2, \ldots, 12$ as given in Table 1. The emergency core cooling system (ECCS) is modeled by a fault tree with 15 basic events (unavailabilities q_j, $j=1, 2, \ldots, 15$) and has the system unavailability $P_i=5.5\times10^{-3}$, the same value P for all segment MLOCAs, $i=1, 2, \ldots, 12$. P is the conditional core damage probability given MLOCA, and the total accident rate is $y=6.27\times10^{-6}$/yr.

The pipe segment MLOCAs are a partitioning of MLOCA in the sense of Section 2.2 with mutually exclusive segments. The criticalities for the initiating events are $CI_i=f_i/f$. These were not obtained in Ref. [12] but are listed here in Table 1. The criticalities of the safety system pipe segments are $CI_j=q_j(\partial P/\partial q_j)/P$ because f cancels out. These are also listed in Table 1. The results show that criticality importances apply consistently for both initiators (process pipes) and unavailabilities (safety system pipes), and the ranking is exactly the same as earlier obtained with DIM. The advantage of criticalities is that they are routinely obtained with most fault tree codes.

The results of this example have clear implications for the importance measures to be used in a two-dimensional matrix concept of planning optimal risk-informed ISI [5]. One dimension should be the segment failure rate f_i for initiators, and the segment unavailability q_j for safety systems. The other dimension should be the Birnbaum measure BI_k in both cases because this makes the combined ranking by the product proportional to the criticality measure.

Remark: Ref. [12] did not calculate the segment initiator criticalities $CI_i=f_i/f$, and it did not divide by the system unavailability $P=5.5\times10^{-3}$ to get the criticalities (FV_j) for unavailabilities (in Table 2 of Ref. [12]), and it did not multiply by the initiating event frequency $f=1.14\times10^{-3}$/yr to get risk-relevant Birnbaum measures. These amendments produced the criticalities in Table 1.

3.2. Roles of the sums of criticality importances

It was shown in Section 2.2 that the sum of criticalities is a relevant importance measure for modules or components that have many single-failure modes. Additional roles of the sums are now pointed out.

The properties and interpretations of criticalities are very important considering what one expects to accomplish with ranking SSCH subject to some PSAA. It is reasonable to expect that a PSAA affects some z_k, and high value of CI_k indicates that this particular z_k should be subject to enhanced PSAA if there is a need to reduce risk or to reallocate resources. High intensity of quality assurance, inspections etc. can be expected to reduce z_k by some reasonable factor. And low frequency of PSAA increases z_k somewhat but does not make a component completely failed, as should happen if RIF_k is taken as a basis for decisions.

Considering an index set S of basic events Z_k subject to some specific PSAA, the impact of a PSAA to the risk can be presented by

Table 1
Ranking of piping segments with DIM and criticality (FV).

Piping segment[a]	Frequency (yr^{-1})	Unavailability	DIM value [12]	Criticality (FV)
EIMHDR		3.00×10^{-3}	0.4812	0.545
EICOREPIP		2.43×10^{-3}	0.3898	0.442
Header 1	1.91×10^{-4}		0.01881	0.168
Header 2	1.91×10^{-4}		0.01881	0.168
Header 3	1.91×10^{-4}		0.01881	0.168
Header 4	1.91×10^{-4}		0.01881	0.168
Feeders-header 4	9.8×10^{-5}		0.01603	0.0860
Feeders-header 1	9.2×10^{-5}		0.009652	0.0807
Feeders-header 2	8.9×10^{-5}		0.009061	0.0781
Feeders-header 3	6.5×10^{-5}		0.008766	0.0570
EICCF1		1.00×10^{-4}	0.007387	0.0182
Coolant channel	1.77×10^{-5}		0.001748	0.0155
Pump-header	6.0×10^{-6}		0.0005909	0.00526
Steam generator-pump	4.0×10^{-6}		0.0003940	0.00351
Header-steam generator	1.0×10^{-6}		9.849×10^{-5}	0.000877
EIRD1		2.21×10^{-3}	1.541×10^{-5}	1.75×10^{-5}
EIRD4		2.21×10^{-3}	1.302×10^{-5}	1.48×10^{-5}
EIRD3		2.21×10^{-3}	1.194×10^{-5}	1.35×10^{-5}
EIRD2		2.21×10^{-3}	1.138×10^{-5}	1.29×10^{-5}
EIPIP3		1.60×10^{-3}	8.608×10^{-6}	9.76×10^{-6}
EIPIP4		1.14×10^{-3}	6.722×10^{-6}	7.62×10^{-6}
EIPIP2		1.11×10^{-3}	6.394×10^{-6}	7.25×10^{-6}
EIPIP1		8.85×10^{-4}	5.532×10^{-6}	6.27×10^{-6}
EICHV1		1.00×10^{-4}	5.781×10^{-7}	6.55×10^{-7}
EICHV2		1.00×10^{-4}	5.375×10^{-7}	6.09×10^{-7}
EICHV4		1.00×10^{-4}	5.104×10^{-7}	5.78×10^{-7}
EICHV3		1.00×10^{-4}	4.778×10^{-7}	5.42×10^{-7}

[a] Safety system components have unavailabilities, MLOCA initiators have frequencies.

Table 2
Example data for component average risk gain calculation.

Component failure mode	Z_k	FV (CI)	RIF	z_k
Valve "A" fails to open	$Z_{A;a}$	0.002	1.7	2.8×10^{-3}
Valve "A" fails to remain closed	$Z_{A;b}$	0.00002	1.1	1.8×10^{-4}
Valve "A" in maintenance	$Z_{A;c}$	0.0035	1.7	4.9×10^{-3}
CCF of valves "A,B,C" to open	$Z_{A,B,C;a}$	0.004	54.0	7.5×10^{-5}
CCF of valves "A" and "B" to open	$Z_{A,B;a}$	0.0007	5.6	1.5×10^{-4}
CCF of valves "A" and "C" to open	$Z_{A,C;a}$	0.0006	4.9	1.5×10^{-4}
Sum of criticalities		0.01082		

a linear model due to modest changes in basic event probabilities:

$$\Delta y=\sum_{k\in S}(\partial y/\partial z_k)\Delta z_k=\sum_{k\in S}\mathrm{BI}_k\Delta z_k=y\sum_{k\in S}\mathrm{CI}_k\frac{\Delta z_k}{z_k}. \tag{8}$$

Often the relative impact to each basic event probability in set S is about the same and the ratios $\Delta z_k/z_k$ are about equal. Then the criticality CI_k indicates the importance of each term and the sum of criticalities is the measure of the total impact.

For example, when S is a subset of basic events that depend on the same failure rate λ, z_k is usually proportional to λ and $\Delta z_k/z_k=\Delta\lambda/\lambda$ is the same for all z_k, $k\in S$. Then $\Delta y/y=\Delta\lambda/\lambda\Sigma_{k\in S}\mathrm{CI}_k$ and the sum of the criticalities indicates the total importance for those events.

Also S could be a group of basic events that belong to the same timing group of periodic testing i.e. the components have to be tested simultaneously or with the same frequency and all z_k, $k\in S$, are proportional to the same test or inspection interval T. Then a change ΔT in T causes changes $\Delta z_k/z_k=\Delta T/T$, which is the same for all $k\in S$, and $\Delta y/y=\Delta T/T\Sigma_{k\in S}\mathrm{CI}_k$. The sum of the criticalities again indicates the total impact on y.

This result is central to balancing and optimization of test and inspection intervals, allowing many low-criticality components testing reduced while a few high-criticality component testing is enhanced to keep the risk unchanged.

3.2.1. Optimization of testing

Optimization of the cost rate per unit time can be written as

$$c(T_1,T_2,\ldots,T_M)=\sum_{m=1}^{M}\frac{C_m}{T_m}+C_A y(T_1,T_2,\ldots,T_M)=\min, \tag{9}$$

where C_A is the cost of an accident, C_m the cost of a group of testing or maintenance actions carried out at regular intervals of T_m. Many components can have the same interval, and components are often grouped together as a *timing group*, for the sake of practical convenience. The accident rate y is a function of the times T_m through the basic event probabilities z_k, each of which is either independent of the times, or proportional to some $T_m(z_k\sim T_m)$, or inversely proportional to some $T_m(z_k\sim 1/T_m)$. One can plot $y(T_m)$ as a function of any single T_m (keeping the others constant) and find the minimum. Generally the minimum of c occurs at larger values of T_m than the minimum of y. It has been shown with the derivatives set to zero [18] that at the optimum the following *cost to risk sensitivity ratio* r_m is the same for all m:

$$r_m=\frac{C_m}{T_m S_m}=C, \text{ the same for all } m, \tag{10}$$

where

$$S_m=T_m\frac{\partial y}{\partial T_m}=T_m\sum_{i=1}^{all}\frac{\partial y}{\partial z_i}\frac{\partial z_i}{\partial T_m}=\sum_{k\in M+}\mathrm{CI}_k-\sum_{j\in M-}\mathrm{CI}_j, \tag{11}$$

where $M+$ is the index set of basic events with the probability proportional to $\sim T_m$, and $M-$ is the index set of the basic events with the probability proportional to $\sim 1/T_m$. Initially the values r_m are not equal, and the idea of the optimization is to increase T_m for the largest r_m and decrease T_m for the smallest r_m. Continuing this iteration leads to the optimum with equal $r_m=C_A$. This procedure was developed early and it allows also non-monotonic $y(T_m)$ [18].

If some z_k is proportional to a power n of T_m higher than 1, this simply means that the corresponding CI_k in the sum S_m is multiplied by n.

It is also important to notice that the ratio of C_m/S_m usually remains the same if an event, like a pipeline break, is subdivided into several events: both the cost and the criticality are subdivided, and the *optimum does not depend on how many events are defined*.

If the above free optimization leads to a risk y above some acceptable limit y_{max}, then one can reduce some T_ms until the risk criterion $y \le y_{max}$ is satisfied. After that one continues constrained optimization changing T_ms so that y remains constant but the cost decreases: increasing T_i for the largest r_i and decreasing z_k for the smallest r_k so that $\Delta y=0$

$$\frac{S_i}{T_i}\Delta T_i = -\frac{S_k}{T_k}\Delta T_k. \tag{12}$$

At the end all r_m are equal to some value C (a virtual accident cost, $C > C_A$). For numerical illustrations, see Ref. [18].

Taking into account the universality, proportionality, additivity, rational interpretations and easy calculation, criticality importance measure seems to be the best for ranking SSCH for PSAA. Naturally, one should include in each application only those failure modes (basic events) of SSCH that are influenced by the specific PSAA being considered. For example, human error events should not be included when ranking basic events for QA or preventive maintenance intensity. However, specific human activities should be focused on and prioritized when training or written procedures can be improved.

4. Configuration management

The problem of configuration management arises in risk monitoring when a plant has one or more failures, possibly complicated by uncertainties about which specific failure modes or common cause failures have occurred. Then one needs to quantify the risk and decide how long the situation can be tolerated, and what to repair or do next. Similar is the case when setting allowed outage (repair) times (AOT) for components or groups of components in advance. Another similar situation occurs when planning future operations with some components to be taken out of service. One needs to quantify the risk for alternative schedules and overlapping failure combinations to select an acceptable course of actions. It is useful to identify sensible importance measures to make the comparisons and assessments without many complete recalculations with the risk model. Essential for these situations is the knowledge (or a plan) that certain components are unavailable or taken out of service. RIF is a useful measure, but it has been defined originally only for basic events. Some new definitions are needed. First, different kinds of configurations are defined as follows:

- *Elementary configuration*: A simultaneous combination of failed and unfailed states of specific basic events, a Boolean product like $X=Z_1Z_2'Z_3$.
- *General configuration*: A set of elementary configurations X_n, each known to be true with some fractional probability p_n; the set of pairs $\{p_n, X_n\}$defines a configuration. This may be called also as an *uncertain* configuration.
- *Momentary configuration*: A configuration (elementary or general) observed, known or possible at a certain time, e.g. right after some tests.
- *Predictive configuration*: A configuration that can be assumed to prevail for some time when making decisions about future actions or priorities.

Momentary configurations can include success states observed at some moment (or in a test). Predictive decision configurations can include success states that can really be guaranteed for the predicted future. Unlike a failure, an observed momentary success state cannot be assumed to generally hold for the future. Success just means a momentary value 0 for z_k in a truly time-dependent analysis. Nominal values z_k are generally used in predictions for a success state that cannot be guaranteed.

Remark: Current risk monitors do not generally calculate the instantaneous time-dependent risk, which is a saw-tooth type nearly periodic function determined by the periodic testing programs for safety components. Current monitors calculate an estimated average risk over some period under a specified configuration. The average calculation is not accurate. In a truly time-dependent risk analysis a momentary success state Z_k' (that is not guaranteed for the future) is an initial value 0 for z_k at the moment of observation, and needs to be replaced with a time-dependent function for z_k, e.g. $z_k=\lambda t$, where λ is a failure rate and t is the time since the observed success state. For a longer horizon the nominal value z_k may be used for $P(Z_k)$ in predictions, which means that Z_k' does not need to appear in the predictive configuration. A detailed time-dependent analysis is outside of the scope of this paper. For an early illustration of time-dependent PSA see Ref. [19].

However, sometimes it is possible to guarantee a success state e.g. by locking a certain valve or switch in the right position, or to conclude some basic event to be impossible (e.g. because of the time of the year or weather conditions), or to know that certain preventive maintenance will not be carried out during the planning horizon. Then a success state can be included in a predictive configuration.

A generalized RIF called risk gain (RG) is defined for elementary configurations in Section 4.1. It is shown how RG can be applied and how it can be determined without negations even when success states are involved. Risk gains for general configurations are defined in Section 4.2, and several alternatives are compared and shown to be equal in certain special cases. In general cases a weighted average method yields most credible values. A real example with common cause failures is quantified. Economically optimal selection of plant modification option is briefly introduced in Section 4.3.

4.1. Risk gain for elementary configurations

A risk gain RG is introduced now as a generalized RIF for a general elementary configuration X based on the conditional risk $y_X=P(Y|X=\text{TRUE})$, namely $\text{RG}(X)=y_X/y$

$$RG(X) = P(Y|X)/P(Y) = \frac{P(XY)}{P(Y)P(X)}. \tag{13}$$

It is possible that $\text{RG} < 1$ when one or more success states are present in X. For a single basic event $\text{RG}_k \equiv \text{RG}(Z_k)=\text{RIF}_k$, and $\text{RG}(Z_k')=y_k^-/y=1-\text{CI}_k$.

RG(X) can be determined for an elementary configuration in two ways:

1. Solving $y_X=P(Y|X)$ from the fault tree Y after setting each event Z_k of X to TRUE and every Z_k' to FALSE according to the observed or predicted state.
2. Solving the fault tree Y for $P(Y)$ and two AND-gates, X and XY, to obtain $P(X)$ and $P(XY)$ to be used in Eq. (13).

A typical case when the risk gain of a configuration can be applied is when a component cannot be repaired while the plant is operating, and there is an incentive to postpone repair exceptionally by some time h. This incurs the following additional risk

compared to the risk over a cycle L [if $\mathrm{RG}(X)>1$]

$$\Delta R=\frac{h}{L}[\mathrm{RG}(X)-1]. \tag{14}$$

Acceptable values for ΔR are generally 1–5%. The same Eq. (14) holds for a regular additional relative risk if some components will be taken regularly out of service for time h with intervals L. If several such activities are planned, a limit is set to the sum of such risk increments.

When a failed unit can be repaired while the plant is operating, the repair can take longer than allowed by the allowed outage time (AOT) defined in technical specifications. The decision about whether to shut down when AOT is reached or to continue at full power for a period h can be made by comparing the accumulated risk $\Delta R=h(y_X-y)=h[\mathrm{RG}(X)-1]y$ to the annual risk or, better yet, to the risk associated with shutdown episode while the failure configuration X is TRUE during the episode. The risk-impact and acceptability of any temporary configuration with increased risk can be assessed by these principles. A practical balancing procedure for planning optimal AOTs for basic events is described in Appendix A.

RG is also useful in prioritizing repairs when more than one component are failed simultaneously and when repair resources are limited. If two components are failed, it is obvious even without knowing $\mathrm{RG}(Z_1Z_2)$ that one should repair first the component with a larger RIF_k, because in this way after the repair the risk is determined by the smaller RIF_k.

4.1.1. Risk gain for elementary configurations with success states

Even in case of success states, negations are not needed in fault trees when solving RG for a configuration. One can develop $\mathrm{RG}(Z_1Z_2')=P(YZ_1Z_2')/[P(Y)P(Z_1Z_2')]=[P(YZ_1)-P(YZ_1Z_2)]/P(Y)[P(Z_1)-P(Z_1Z_2)]$. Applying this principle yields for independent events

$$\mathrm{RG}(Z_1Z_2')=\frac{\mathrm{RIF}(Z_1)-z_2\mathrm{RG}(Z_1Z_2)}{1-z_2}, \tag{15}$$

$$\mathrm{RG}(Z_1'Z_2')=\frac{1-z_1\mathrm{RIF}(Z_1)-z_2\mathrm{RIF}(Z_2)+z_1z_2\mathrm{RG}(Z_1Z_2)}{(1-z_1)(1-z_2)}, \tag{16}$$

$$\mathrm{RG}(Z_1Z_2Z_3')=\frac{\mathrm{RG}(Z_1Z_2)-z_3\mathrm{RG}(Z_1Z_2Z_3)}{1-z_3}. \tag{17}$$

In general $\mathrm{RG}(Z_1Z_2)$ cannot be presented in terms of $\mathrm{RIF}(Z_1)$ and $\mathrm{RIF}(Z_2)$. In some special case this is possible. If Z_1 and Z_2 never appear together in a MCS and do not have similar MCSs (in the sense of Section 2.2), then $\mathrm{RG}(Z_1Z_2)=\mathrm{RIF}(Z_1)+\mathrm{RIF}(Z_2)-1$. If Z_1 and Z_2 are m.e., e.g. failure modes of the same component, then $\mathrm{RG}(Z_1'Z_2)=\mathrm{RIF}(Z_2)$ and the success of Z_1 has no value if Z_2 is failed.

The results (15)–(17) are particularly handy when CCFs are explicitly modeled because then many risk gains are obtained as RIF-values for CCF. This is illustrated with an example in Section 4.2.1.

4.2. Risk gain for general configurations

Consider a set of N elementary configurations X_n possible based on the information available ($n=1, 2, \ldots, N$). The following three alternative methods are tested in numerical examples.

1) *Conditional output method* (*COM*). In this method the Boolean union of elementary configurations, $X=X_1+\ldots+X_N$, is defined as general configuration and a condition for Y. The risk gain is defined as in Eq. (13)

$$\mathrm{RG}_{\mathrm{COM}}(X)=\frac{P(XY)}{P(X)P(Y)}. \tag{18}$$

2) *Conditional input method* (*CIM*). In CIM the basic event probabilities z_k in y are replaced by the conditional values $P(Z_k|X)=P(Z_kX)/P(X)$. The resulting y_X defines the risk gain

$$\mathrm{RG}_{\mathrm{CIM}}(X)=y_X/y. \tag{19}$$

3) *The weighted average method* (*WAM*). The general configuration is defined by pairs (p_n, X_n) where each X_n is one of the elementary configurations that can be TRUE, with probability p_n ,which depends on all knowledge at the time of the decision, not necessarily only on the basic a priori probability $P(X_n)$. The conditional risks are $y_n=P(Y|X_n)=P(YX_n)/P(X_n)$, the individual risk gains are $\mathrm{RG}(X_n)=y_n/y$, and the overall configuration risk gain is the weighted average

$$\mathrm{RG}_{\mathrm{WAM}}(\{p_n,X_n\})=\frac{\sum_{n=1}^{N}p_nP(Y|X_n)}{\left(\sum_{n=1}^{N}p_n\right)P(Y)}=\frac{\sum_{n=1}^{N}p_nP(YX_n)}{\left(\sum_{n=1}^{N}p_n\right)P(Y)P(X_n)}. \tag{20}$$

In case of a single elementary configuration X all three methods yield identical results. The identity of Eqs. (18) and (20) is obvious when $N=1$. CIM would replace z_k of each event that is a factor in X by 1, leaving other input probabilities unchanged. This yields the same result as WAM and COM. In some other cases the methods also agree, but not always.

Theorem 1. *WAM, COM and CIM are identical when multiple elementary configurations X_n are mutually exclusive but are independent of all other basic events, and the configurations X_n appear as such in minimal cut sets. In this case $p_n=P(X_n)$, $n=1, 2, \ldots, N$.*

The conditions of this theorem are satisfied e.g. when all X_n are m.e. single basic events that are independent of all other basic events. To prove Theorem 1 one may begin with the expression

$$Y=\sum_{n=1}^{N}X_nG_n+H, \tag{21}$$

X_nG_n is the union of the minimal cut sets that contain X_n as a factor; H is the union of the minimal cut sets that contain no X_n and no basic event of any X_n.

Using the assumed properties and developing the conditional probabilities one can show that COM, WAM and CIM yield the same risk gain, with $p_n=P(X_n)$, $n=1,2, \ldots, N$.

However, the methods are not identical for all kinds of configurations. This can be shown with examples.

Example 1. Consider a parallel system of three independent identical 100% components. The system logic is such that all three components must fail for the system to fail. The system TOP event is $Y=Z_1Z_2Z_3$ with numerical values $P(Z_k)=z_k=0.1$, $k=1,2,3$. The unconditional risk is $y=z_1z_2z_3=0.001$. The configuration information at the time of observation is: component 1 or component 2 is down, but not both. (This kind of information can be based on partial inspection and the output capacity of the system, for example).

Based on what is known, it is obvious that the momentary risk is zero because the TOP event is not TRUE, and the predicted risk without success assumption must be the same as if one knows exactly component 1 (or component 2) to be failed, i.e. $y_X=z_2z_3=0.01$. Then the correct risk gain is $\mathrm{RG}=y_X/y=10$. Let us see what different methods can tell about the situation.

COM: The *momentary* configuration as a Boolean function is $X=Z_1Z_2+Z_1'Z_2$. Because of the success states this method yields the new risk $y_X=P(Y|X)=0$ and $\mathrm{RG}_{\mathrm{COM}}(X)=0$.

CIM: With $X=Z_1Z_2'+Z_1'Z_2$ the conditional input probabilities are $z_{1c}=P(Z_1X)/P(X)=z_1(1-z_2)/(z_1+z_2-2z_1z_2)=0.5$, $z_{2c}=0.5$, and $z_3=0.1$. This method yields the risk $y_{\mathrm{CIM}}=z_{1c}z_{2c}z_{3c}=0.025$, and $\mathrm{RG}_{\mathrm{CIM}}=25$.

WAM: Both elementary configurations Z_1Z_2' and $Z_1'Z_2$ yield the conditional $y=0$, and therefore $RG_{WAM}=0$. In this case COM and WAM agree and must be right because one component is unfailed.

Let us now take the predictive viewpoint that success states are not assumed for the future horizon. Only failed states are assumed, and it is known in this example that Z_1Z_2 is not one of the allowed configurations. In this example the elementary configuration $X_1=Z_1$ holds for the future with probability $p_1=z_1(1-z_2)=0.09$ and the elementary configuration $X_2=Z_2$ with probability $p_2=z_2(1-z_1)=0.09$. For each elementary configuration the risk gains is $RG(X_1)=RG(X_2)=10$. Weighting the corresponding risks and gain factors we have the averaged predictions $y_{WAM}=[p_1P(Y|X_1)+p_2P(Y|X_2)]/(p_1+p_2)=0.01$ and $RG_{WAM}=[p_1RG(X_1)+p_2RG(X_2)]/(p_1+p_2)=10$. This is exactly what is expected based on the information that exactly one component is failed. Please observe that even when $X_1=Z_1$, p_1 depends on both z_1 and z_2. This is because the observation (the knowledge that both Z_1 and Z_2 are not down) is taken into account in the probability p_n even when Z_2' or Z_1' are not in the predictive configurations X_n. For COM and CIM the union of alternative states means configuration $X=Z_1+Z_2$, and these methods yield $RG_{COM}(X)=5.263$ and $RG_{CIM}(X)=27.68$. Neither one agrees with $RG_{WAM}=10$.

Example 2. Consider the same system and data as in Example 1, with different momentary knowledge: at the time of decision it is known that component 1 or 2 or both are down. For COM and CIM this means predictive configuration $X=Z_1+Z_2+Z_1Z_2=Z_1+Z_2$ based on Boolean algebra. These methods yield $RG_{COM}(X)=5.263$ and $RG_{CIM}(X)=27.68$ as obtained for the predictive configuration in Example 1. For WAM there are three elementary configurations, Z_1, Z_2 and Z_1Z_2 with probabilities $p_1=z_1(1-z_2)$, $p_2=z_2(1-z_1)$ and $p_3=z_1z_2$, respectively. These yield $RG_{WAM}=14.74$. This is consistent with the fact that more failure combinations (configurations) are possible now than in Example 1, and the true RG must be larger than 10. CIM and COM do not show any difference compared to Example 1.

Conclusion: When multiple elementary configurations are possible in a decision making situation, the conditional fault tree input method and the conditional fault tree output method are not generally valid for calculating the changed risk and risk gain. A single Boolean union cannot generally describe correctly the alternatives of what is known momentarily and what is to be assumed for the future. Neither COM nor CIM is generally consistent with WAM.

From now on WAM is considered a valid method for assessing uncertain configurations. The following examples illustrate how available information impacts results.

Example 3. Consider a redundant $2\times100\%$ system with $Y=Z_1Z_2+Z_{1,2}$, where Z_1 and Z_2 are single failures of components 1 and 2, respectively, and $Z_{1,2}$ is a CCF failing both components simultaneously. The events are assumed to be mutually independent and the numerical unavailabilities are $z_1=z_2=0.10$ and $z_{1,2}=0.01$, which yield $y=z_1z_2+z_{1,2}-z_1z_2z_{1,2}=0.0199$. Several combinations of momentary knowledge and predictive information can be considered:

1) Single failure Z_1 is observed in a test, while component 2 is unfailed. (This implies that neither Z_2 nor $Z_{1,2}$ have occurred).
   a) If the success states could be guaranteed for near future, then no risk would be involved, because with the only elementary configuration $X=Z_1Z_2'Z_{1,2}'$ Eq. (13) yields $RG(Z_1'Z_2'Z_{1,2}')=0$.
   b) If one can guarantee the success Z_2 but not $Z_{1,2}'$, there is only one prediction configuration $X=Z_1Z_2'$, and $RG(Z_1Z_2')=z_{1,2}/y=0.5025$.
   c) If one can guarantee success $Z_{1,2}'$ but not Z_2', then $RG(Z_1Z_{1,2}')=z_2/y=5.025$.
   d) If one cannot guarantee any success states, $RG(Z_1)=RIF(Z_1)=(z_2+z_{1,2}-z_2z_{1,2})/y=5.477$.
2) It is known that either component 1 or component 2 is failed, but not both.
   The momentary configuration is either $Z_1Z_2'Z_{1,2}'$ or $Z_1'Z_2Z_{1,2}'$. If success states could be guaranteed, no added risk would be involved and $RG(X)=0$. When no successes can be guaranteed, the prediction configurations are $X_1=Z_1$ and $X_2=Z_2$. The configuration probabilities must take into account the actual momentary information, so $p_1=z_1(1-z_2)(1-z_{1,2})$ and $p_2=z_2(1-z_1)(1-z_{1,2})$. Eq. (20) yields $RG_{WAM}=5.477$.
3) Both components are known to be failed.
   Both possible configurations Z_1Z_2 and $Z_{1,2}$ have the same risk gain $1/y=50.25$. The weighted average is also $RG_{WAM}=50.25$.
4) Component 1 is known to be failed, but it is not known whether it is due to Z_1 or $Z_{1,2}$, i.e. nothing is known about component 2.
   The momentary and prediction configurations are now $X_1=Z_1$ and $X_2=Z_{1,2}$.
   a) If it is known that only one of the configurations has occurred but not both, then $p_1=z_1(1-z_{1,2})=0.099$ and $p_2=z_{1,2}(1-z_1)=0.009$. Eq. (21) with $N=2$ yields $RG=9.209$.
   b) If it is possible that both events Z_1 and $Z_{1,2}$ have occurred, there is a third configuration $X_3=Z_1Z_{1,2}$ possible with $p_3=Z_1Z_{1,2}=0.001$. With $N=3$ now Eq. (20) yields $RG=9.595$.

4.2.1. A component with common cause failures

Consider a single component A with multiple failure modes, single failure modes a, b, c, etc., and some CCFs that fail this particular component A and some others like B, C, etc., with corresponding failure modes. These failures are assumed to be mutually exclusive. The basic events $Z_{A;a}$, $Z_{A;b}$, $Z_{A;c}$, $Z_{A,B;a}$, $Z_{A,C;a}$, $Z_{A,B,C;a}$ etc. are all associated with A (A is one of the sub-indices). There are some other mutual exclusivities due to symmetry: $Z_{A,B,a}$ and $Z_{A,B,C;a}$ are m.e. with $Z_{B;a}$, while $Z_{A,C;a}$ and $Z_{A,B,C;a}$ are m.e. with $Z_{C;a}$. But all these events are assumed to be m.s.i. with all other components and events that do not share a common component and failure mode index. Then the set of events associated with A satisfies the characteristics of the set J defined in Section 2.1, and the corresponding relationships between the importance measures, e.g. Eq. (5).

Assume now that the only momentary information available is that component A is failed. One of the events with A as a subindex has occurred, but we do not know which one. How much higher is the risk in this situation, i.e. what is the risk gain based on the information available? One can prove with similar principles as Theorem 1.

Theorem 2. *WAM, COM and CIM are identical for the union of mutually exclusive basic events Z_k associated with a single component and satisfying the rules of set J. The elementary configurations are $Z_k\in J$ with probabilities $p_k=P(Z_k)$, k in index set J. The risk gain can be obtained as the weighted sum of RIF-values of the basic events*

$$RG_{WAM}=\frac{\sum_{k\in J}P(Z_k)RIF(Z_k)}{\sum_{k\in J}P(Z_k)}. \tag{22}$$

Furthermore, because of the relationship (5) one can show the following:

Theorem 3. *For the single component basic events defined in Theorem 2, the following balancing method expression (BAM) yields*

the same risk gain as WAM, COM and CIM

$$\mathrm{RG_{BAM}} = 1 + \frac{\sum_{k\in J}\mathrm{CI}(Z_k)}{\sum_{k\in J}P(Z_k)} - \sum_{j\in J}\mathrm{CI}(Z_j). \tag{23}$$

Thus, in this case the risk gain can be obtained using criticalities in Eq. (23) or RIFs of the basic events as in Eq. (22).

To demonstrate the risk gain quantification in a real example for three valves *A*, *B* and *C* with common cause failures, data is presented in Table 2 [20]. Eq. (5) was used to solve the probability values z_k from the importance values. The sub-indices *A*, *B*, *C* indicate, which components are failed by each event *Z*, and a, b and c indicate failure modes "fail to open", "fail to remain closed" and "undergoing maintenance", respectively. All listed events fail component *A* and satisfy the conditions of set *J* as it was defined in Section 2.

Using Eq. (22) for all six events yields the risk gain of component A failure (for a completely unknown cause), $\mathrm{RG}(A)=\mathbf{2.29}$. Earlier various ad hoc methods have given component-RIF values from 1.7 through 69 [20].

If one knows after a test that valve *A failed to open*, one knows that one of the four "fail to open" events is TRUE, but one does not know exactly which one. Then the valve "*A*" risk gain is the weighted average of the RIF values of these four events (without $Z_{A;b}$ and $Z_{A;c}$). This value is $\mathrm{RG_{WAM}}$ (*A* fails to open)=3.27. If one assumes all basic events to be m.s.i., the results are not much different because the basic event probabilities are small.

If a test has revealed that valves *A* and *B* failed to open while *C* opened, possible predictive elementary configurations are $Z_{A;a}\,Z_{B;a}$ and $Z_{A,B;a}$. Because these have the same impact on the system, $\mathrm{RG}(Z_{A;a}Z_{B;a})$ must be equal to $\mathrm{RIF}(Z_{A,B;a})=5.6$. This is the predictive risk gain for the situation. By the way, the balancing Eq. (23) yields for this combination $\mathrm{RG_{BAM}}(Z_{A;a}Z_{B;a})=1.7$.

One could improve the situation by deciding not to carry out maintenance for valve *C* until repairs of *A* or *B* take place. Then $Z_{C;c}'$ becomes a multiplier to configurations $Z_{A;a}Z_{B;a}$ and $Z_{A,B;a}$. Eqs. (15) and (17) show that the risk gains are reduced. Taking into account $\mathrm{RG}(Z_{A;a}Z_{B;a}Z_{C;c})=\mathrm{RG}(Z_{A,B;a}Z_{C;c})=\mathrm{RIF}(Z_{A,B,C;a})=54$ and assuming symmetry $z_{C;c}=z_{A;c}=0.0049$, yield RG=5.36.

Remark: Kim et al. [21] compared the balancing method with the conditional input method in several examples with CCF, and obtained similar results with both methods. This is a natural consequence from Theorem 3, and proves nothing about more general validity of CIM or BAM.

Vasseur et al. [22] compared COM (calling it "fictive fault tree") with average RIF in some common cause failure examples. The agreement observed in their examples follows from Theorem 2, without proving general validity of COM.

As was concluded in Section 3, criticality importance is relevant for ranking components for quality assurance, testing programs or other PSAA. A test interval, for example, generally influences all hardware failure modes of a component. Based on proportionality and reasonable additivity, the appropriate criticality for component *A* is the sum of all hardware failure modes, $\mathrm{CI}(A)=0.0073$. For an activity that is intended to reduce CCF contributions, the relevant criticality is the sum over all CCFs for the component group, in this case 0.0059. These values then ought to be compared to similar criticalities of other components and component groups.

PSAA is normally applied to all similar components in a system and influences similar basic events of such a component group. As discussed in Section 3.2 the additive property of CI justifies using the sum of PSAA-relevant criticalities as a basis for ranking a component or a CCF-group. Considering the hardware criticalities as relevant in this case we obtain $\mathrm{CI}(A,B,C|\text{hardware})=\mathrm{CI}(Z_{A;a})+\mathrm{CI}(Z_{A;b})+\mathrm{CI}(Z_{B;a})+\mathrm{CI}(Z_{B;b})+\mathrm{CI}(Z_{C;a})+\mathrm{CI}(Z_{C;b})+\mathrm{CI}(Z_{A,B;a})+\mathrm{CI}(Z_{A,C;a})+\mathrm{CI}(Z_{B,C;a})+\mathrm{CI}(Z_{A,B,C;a})=0.0120$. In economic considerations the cost of PSAA then has to include the cost of actions for all three components.

4.3. Sustainable plant improvement

If there is a plan to improve safety by implementing a plant change identified by index ν, with the present value investment cost C_ν that results in risk level y_ν, there is an economic requirement $C_\nu + aC_A y_\nu < aC_A y$, where *a* is the discount factor, that must be satisfied to justify the plant modification. Typical values indicate that reduction of a major accident ($C_A=10^9$ €) rate by 10^{-6} per year may justify expenditures 2×10^4 €, depending also on the remaining lifetime of the plant.

If there are many options and combinations of modifications, one has to consider the present values and risks of all options and select the one with the smallest sum $C_\nu + aC_A y_\nu$, including the "no change" option with cost $aC_A y$. In this application importance measures may not play a role except when the risk-impact of a plant modification can be presented by some configuration X_ν so that $y_\nu=\mathrm{RG}(X_\nu)y$. It has been shown with an application to flood risk reduction options that the discounted cost criterion leads to a different optimal solution than the simple benefit/cost ratio $(y-y_\nu)/C_\nu$ or when simply minimizing the product $y_\nu C_\nu$ [1].

5. Summary and conclusions

Roles, extensions and applications of various importance measures in risk-informed decision making were described. Basic measures and their relationships were pointed out for independent and mutually exclusive events and for groups of events associated with common cause failures. The roles of importance measures were described manly in two groups of activities: (a) ranking safety significance of systems, structures, components and human actions (SSCH) for preventive safety assurance activities (PSAA) which typically include planning of safety classification, testing and inspection programs, quality assurance, preventive maintenance, training, procedures and human factors development, and (b) making decisions about permissible permanent and temporary configurations and allowed configuration times for regulation, for planning technical specifications and for on-line risk monitoring. Identified characteristics of criticality importance make it an appropriate measure for ranking as it indicates both component sensitivity to PSAA and its impact on risk. Criticality and sums of criticalities have advantages also for optimization based on risk and cost. It was demonstrated that criticality gives consistent ranking for both initiating events and safety system components in in-service pipe inspection ranking as an example. Ranking based on criticalities typically leads to reduced total burden in a preventive activity while the risk is reduced or maintained constant. In applications one has to select subsets of basic events that are influenced by the PSAA being applied. For other types of applications one may consider other kinds of multi-event importances like the network importance [26] that takes into account capacities of loads in power systems.

Risk increase factor (RIF) is most appropriately used to assess corrective priorities and acceptability of risk in a situation when a component is already known to be failed, or when planning to take a component out of service for maintenance. Precise definitions were introduced for elementary and general multi-failure configurations and it was shown how they can be assessed under uncertainties, in particular when common cause failures are involved. A general weighted average method was compared to several other candidate methods in benchmark cases. Conditions were found under which several methods yield equal results. The weighting method is based on the definition and calculation of

risk gain (RG), a generalized RIF, when a momentary configuration is known or only known with probabilities. It was shown how momentary observations and predictive configurations can differ from each other, especially in case of success states, and how the specific momentary knowledge should be taken into account. The risk gain can also be less than 1 when success states can be justified. Relationships of RG to conventional importance measures were described. Generally RG can be obtained using fault tree techniques and in some cases directly from basic event importances. An example balancing of allowed outage times demonstrated that many AOTs may be extended when a few are shortened while the risk is reduced or remains unchanged. Also in configuration control it is important to select subsets of basic events that actually can be causes for the configuration, or are determined by the decision maker. Explicit modeling of common cause failures seems to make application of importance measures rather straightforward. When planning plant modifications, sustainable development requires that both risk and economy are taken into account in decision making. Basic methods for this purpose were pointed out.

Summary conclusions:

- Mutual exclusivity assumptions change relationships between importance measures compared to independent events.
- Criticality importance is the measure for ranking SSCH for preventive activities because of sufficient linearity, additivity and capability to rank consistently initiators and safety components. BI or RIF alone are not recommended for ranking.
- Subsets of components and basic events should be selected for ranking so that they are sensitive to the specific type of PSAA being considered; all unavailabilities are not influenced by all PSAA.
- RIF and RG are relevant measures for configuration management when combined with the weighted average method (WAM), applicable to complex and uncertain configurations.
- Predictive decision configurations must be identified separately from the momentary observed configurations, and the weighting probabilities are based on all knowledge at hand in addition to the inherent properties and probabilities of the events involved.
- The basic events to be included in the configurations are not necessarily all events of a component; only some of them may be possible causes of a configuration, depending on what has been observed.
- Other methods COM, CIM and BAM yield correct results in special cases; conditions for this have been identified.
- It is advantageous to model CCFs explicitly as basic events because then criticalities and risk gains can be obtained directly by standard fault tree codes, often from the basic importance measures.

Appendix A. Acceptance and balancing of AOTs

Eq. (14) applies also for any single component or basic event that is observed in a failed state. It is customary to specify in advance and document in technical specifications allowed outage times such that an event (single failure or CCF) has to be repaired within a specified AOT, otherwise the plant must be shut down or the risk must be compensated by other means. Then $h=$AOT and RG equals RIF of the basic event to which AOT is assigned. The same or different AOT may be assigned to the durations of preventive maintenance actions that take a component out of service. The initial values for these AOTs are determined by engineering practice to 3–5 times the mean repair time, or they may be determined so that each term (14) is a small fraction of the annual risk, e.g. 10^{-4}. At the end it is possible to verify with the risk model that the maximum total impact of the selected AOTs is acceptable. However, this approach does not take into account that some events occur more frequently than others and therefore perhaps contribute more to the time-average (annual) risk. They should have shorter AOTs than those events that occur infrequently. This leads to the idea of balancing AOTs so that each has about the same impact on the annual risk, and the total risk remains the same or lower. This approach means that the following relative risk contributions are quantified

$$\Delta F_k = [\mathrm{RIF}(Z_k)-1]\Phi_k \mathrm{AOT}_k, \tag{A.1}$$

where Φ_k is the expected frequency of all AOT-situations for event Z_k with the preliminary value AOT_k. The events include single failures and CCF-events failing multiple components, and even scheduled maintenance outages during plant operation. Then the average ΔF_{av} of the values ΔF_{kk} is determined, and

A) for each event with ΔF_k clearly smaller than ΔF_{av}, increase AOT_k to a new value closer to $\mathrm{AOT}_k \cdot \Delta F_{\mathrm{av}}/\Delta F_k$,
B) for each event with ΔF_k clearly larger than ΔF_{av}, shorten AOT_k to a new value closer to $\mathrm{AOT}_k \cdot \Delta F_{\mathrm{av}}/\Delta F_k$.

Because there are usually much more events in group *A* than in group *B*, this procedure leads to increasing a large number of AOTs while reducing a few most important ones. This can have an additional advantage that occurrences of repair times exceeding AOT are reduced, which has an economic advantage and also reduces risks associated with plant shutdown and startup operations. The procedure tends to keep the total risk constant because it keeps the sum of risk increments ΔF_k about unchanged. This should be verified at the end, taking into account that mean repair times can also change slightly when AOT is changed.

This procedure is simpler than some alternatives [23,24], yet test applications have shown quite comparable results. For example, in the example of Ref. [23] with 66 basic events of a low-pressure emergency core cooling system and the original allowed AOT values 0, 8, 24 h, 3 and 21 days, the above procedure led to 29 extensions of AOT to the next category, and only 2 shortenings to the next lower category while the total risk contribution was reduced by a factor of two. Evidently the total burden associated with AOTs is reduced by this procedure even though cost-optimization is not formally employed.

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